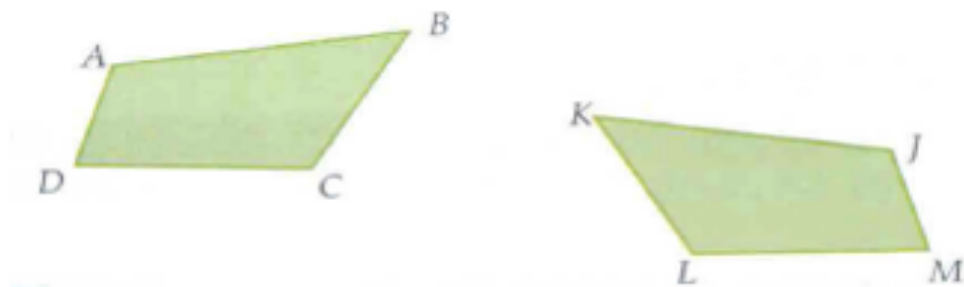


10E Congruent Triangles Part 2

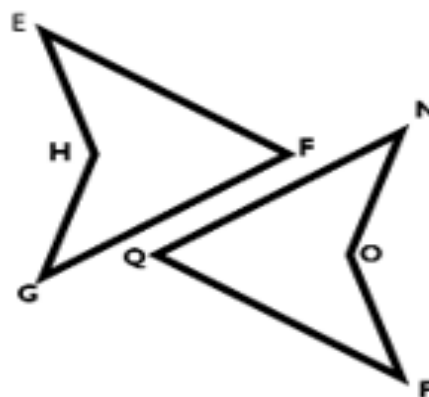
Learning Objective	Identify conditions for congruency and similarity in shapes and create and test algorithms designed to test for congruency and similarity
Today We Are	Determining that triangles are congruent using the four rules, SSS, SAS, AAS, RHS
Indicators of Progress	I can use the four rules for congruent triangles

Warm Up Activity (Glue into your books)

- Identify the matching (corresponding) sides and angles in the following congruent quadrilaterals:



- $\angle A \equiv \angle J$
- $\angle B \equiv \angle K$
- $\angle C \equiv \angle L$
- $\angle D \equiv \angle M$



- e) Line $EF \equiv$ Line NQ
- f) Line $FG \equiv$ Line QP
- g) Line $GH \equiv$ Line PO
- h) Line $HE \equiv$ Line ON



side-side-side

side-angle-side



Angle-angle-side

Right angle-Hypotenuse-side



SSS



RHS

SAS

AAS

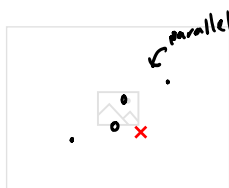


AAS

RHS

c) Write a proof for these pairs of congruent triangles

SSS **AAS**
RHS SAS



$\triangle ABC \equiv \triangle EDC$ (AAS)

$AC = EC$

$\angle ABC = \angle EDC$ (alternate angles)

$\angle ACB = \angle ECD$ (vertically opposite angles)



